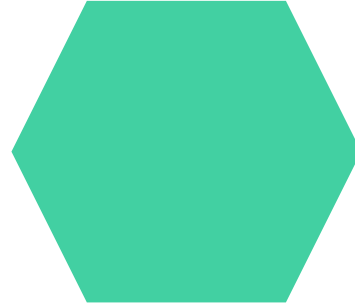
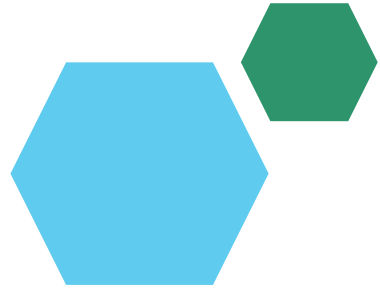




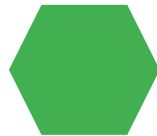
Project Title - Seasonal Agriculture Performance Analysis

Student Name :- Soumyadeep Masanta

College name :- College of Engineering and Management, Kolaghat

AICTE STU ID :- STU6a132fe9011381779642345

Internship ID :- INTERNSHIP_17830691666a4779eecfe8a



PROBLEM STATEMENT



Agricultural performance is shaped by seasonal shifts in weather, resource use, and market conditions — but raw farm-level data doesn't on its own reveal how outcomes actually change from one season to another. This project analyzes a seasonal agriculture dataset (covering Kharif, Rabi, and Zaid seasons across multiple states and crops) to uncover meaningful patterns, trends, and statistically significant differences in yield, profitability, resource usage, and risk across seasons — and to translate those findings into evidence-based recommendations for better seasonal planning.



Project Details :

This project performs an end-to-end data analytics study on a seasonal agriculture performance dataset covering farms across multiple Indian states, crops, and irrigation methods over three seasons – Kharif, Rabi, and Zaid. The analysis was carried out in Python (Google Colab) using Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, and Scikit-learn.

The workflow included:

- Data Cleaning & Preparation - handling missing values (season-wise median imputation), removing duplicates, and treating outliers using the IQR (winsorization) method.
- Exploratory Data Analysis (EDA) - visualizing distribution of records across seasons, states, and crops, and comparing yield, profit, revenue, and cost across seasons.
- Statistical Testing - one-way ANOVA and Tukey HSD post-hoc tests to check whether seasonal differences in yield, profit, water efficiency, and disease/pest risk are statistically significant; a two-way ANOVA to test the combined effect of Season and Irrigation Method; and a Chi-square test to check whether irrigation method choice depends on season.
- Correlation & Relationship Analysis - identifying which environmental and resource factors (rainfall, soil pH, fertilizer, etc.) are most strongly linked to yield and profit.
- Predictive Modeling - building an OLS regression model (for interpretable coefficients) and a Random Forest model (for feature importance) to identify key drivers of crop yield.
- Clustering - applying K-Means to segment farms into performance groups independent of season, and checking whether these clusters align with seasonal patterns.
- Cost-Benefit & Risk Analysis - computing profit margin %, ROI, and disease/pest risk by season to evaluate not just productivity but profitability and risk exposure.

The key finding is that Kharif is the most profitable season but carries the highest disease/pest risk, Zaid is a loss-making season for most farms, and irrigation method (not season) is the strongest lever farmers can control – since Rainfed/Drip consistently outperform Flood irrigation regardless of season.

WHO ARE THE END USERS?

Farmers & Agricultural Producers - to decide which crops/seasons to prioritize, choose better irrigation methods, and reduce risk exposure (e.g., avoiding Flood irrigation, being cautious about Zaid-season investments).

Agricultural Extension Officers & Advisors - to guide farmers with season-specific, data-backed recommendations rather than generic advice.

State/Regional Agriculture Departments & Policymakers - to design seasonal support schemes, subsidies, or insurance policies (e.g., risk mitigation for Zaid season, incentives for efficient irrigation).

Agri-Finance & Insurance Providers - to assess seasonal risk and profitability patterns when offering crop loans or crop insurance.

Agribusiness & Supply Chain Companies - to plan procurement, storage, and pricing strategies around expected seasonal yield and profit trends.

Researchers & Students - as a reference case study for applying statistical and ML techniques to real-world agricultural data.

Technology Used

Language & Environment

Python 3

Google Colab / Jupyter Notebook

Data Handling

Pandas - data cleaning, wrangling, pivot tables

NumPy - numerical computations

Data Visualization

Matplotlib - charts and plots

Seaborn - statistical visualizations (pairplots, heatmaps, distributions)

Statistical Analysis

SciPy - ANOVA, Chi-square test, skewness

Statsmodels - Tukey HSD post-hoc test, two-way ANOVA, OLS regression

Machine Learning

Scikit-learn - Random Forest (feature importance), K-Means clustering, train-test split, feature scaling

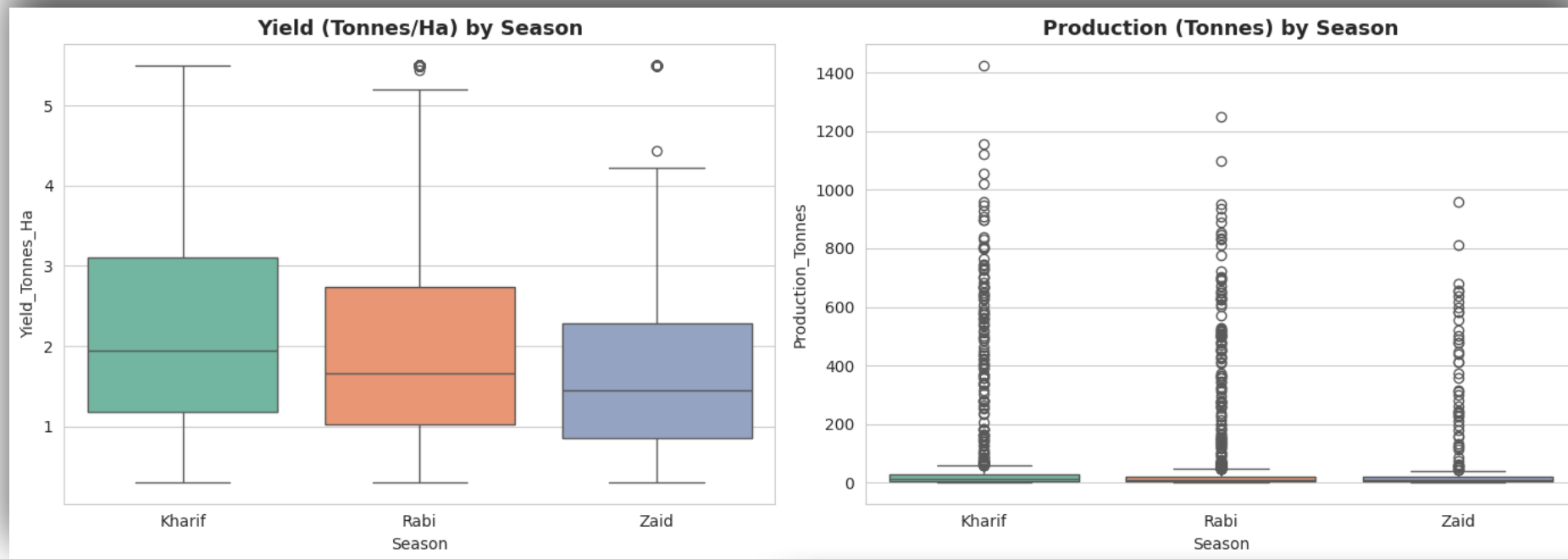
Version Control

Git & GitHub - project versioning and submission



RESULTS

Yield & Production Across Seasons



Kharif shows the highest median yield (~2.0 t/ha), followed by Rabi (~1.7 t/ha) and Zaid (~1.5 t/ha)

Differences between all three seasons are statistically significant (ANOVA, $p < 0.001$)

Zaid also shows the widest spread and more high outliers – less consistent performance

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

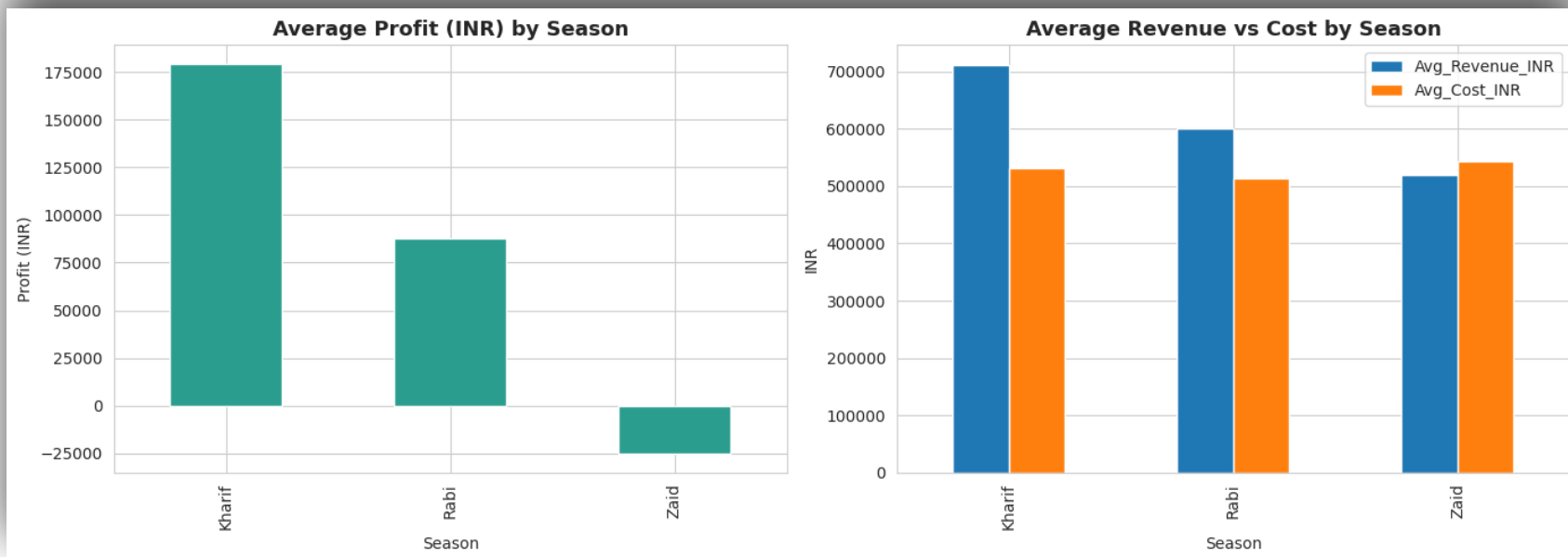
sns.boxplot(data=df, x='Season', y='Yield_Tonnes_Ha', ax=axes[0], palette='Set2')
axes[0].set_title('Yield (Tonnes/Ha) by Season')

sns.boxplot(data=df, x='Season', y='Production_Tonnes', ax=axes[1], palette='Set2')
axes[1].set_title('Production (Tonnes) by Season')

plt.tight_layout()
plt.show()
```

RESULTS

Profitability Across Seasons



Kharif is the most profitable season by far (~₹1.79 lakh average profit)

Rabi is moderately profitable (~₹0.88 lakh)

Zaid runs at an average loss (~₹25,000) – its average cost exceeds its average revenue, unlike the other two seasons

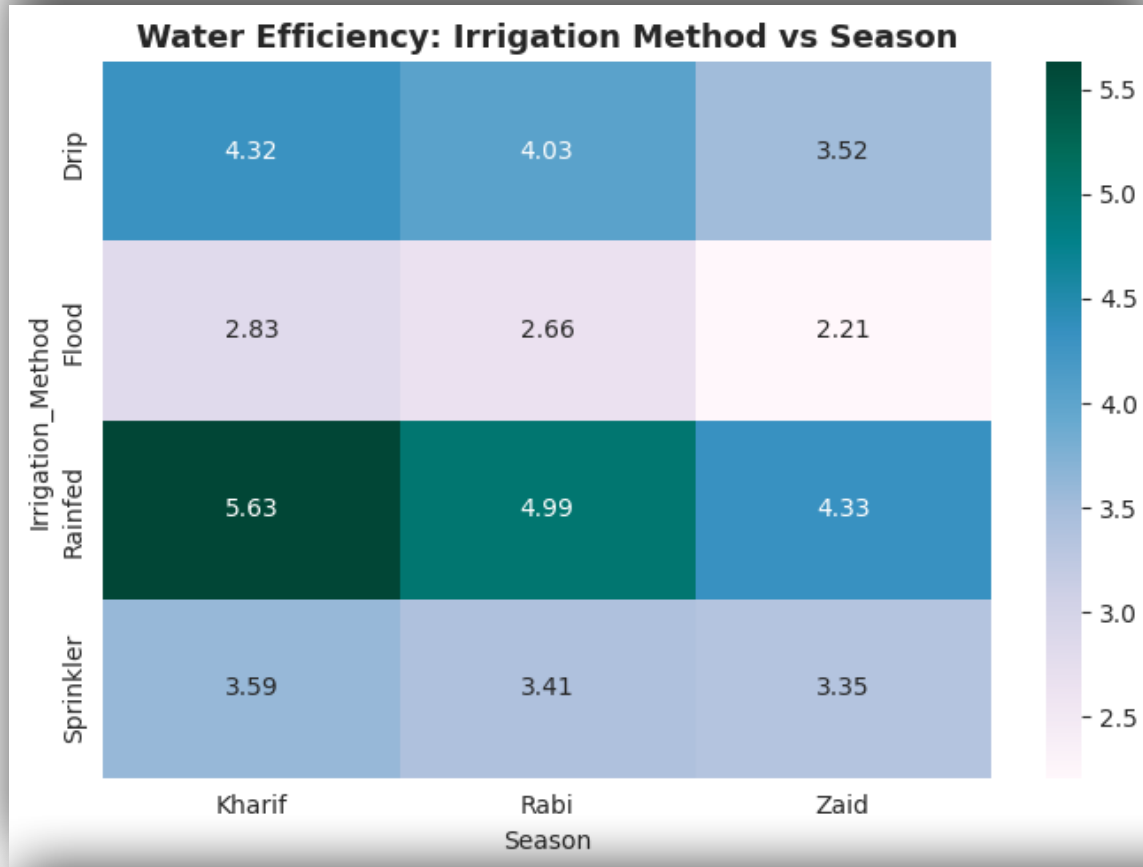
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

season_summary['Avg_Profit_INR'].plot(kind='bar', ax=axes[0], color='#2A9D8F')
axes[0].set_title('Average Profit (INR) by Season')
axes[0].set_ylabel('Profit (INR)')

season_summary[['Avg_Revenue_INR', 'Avg_Cost_INR']].plot(kind='bar', ax=axes[1])
axes[1].set_title('Average Revenue vs Cost by Season')
axes[1].set_ylabel('INR')

plt.tight_layout()
plt.show()
```

RESULTS



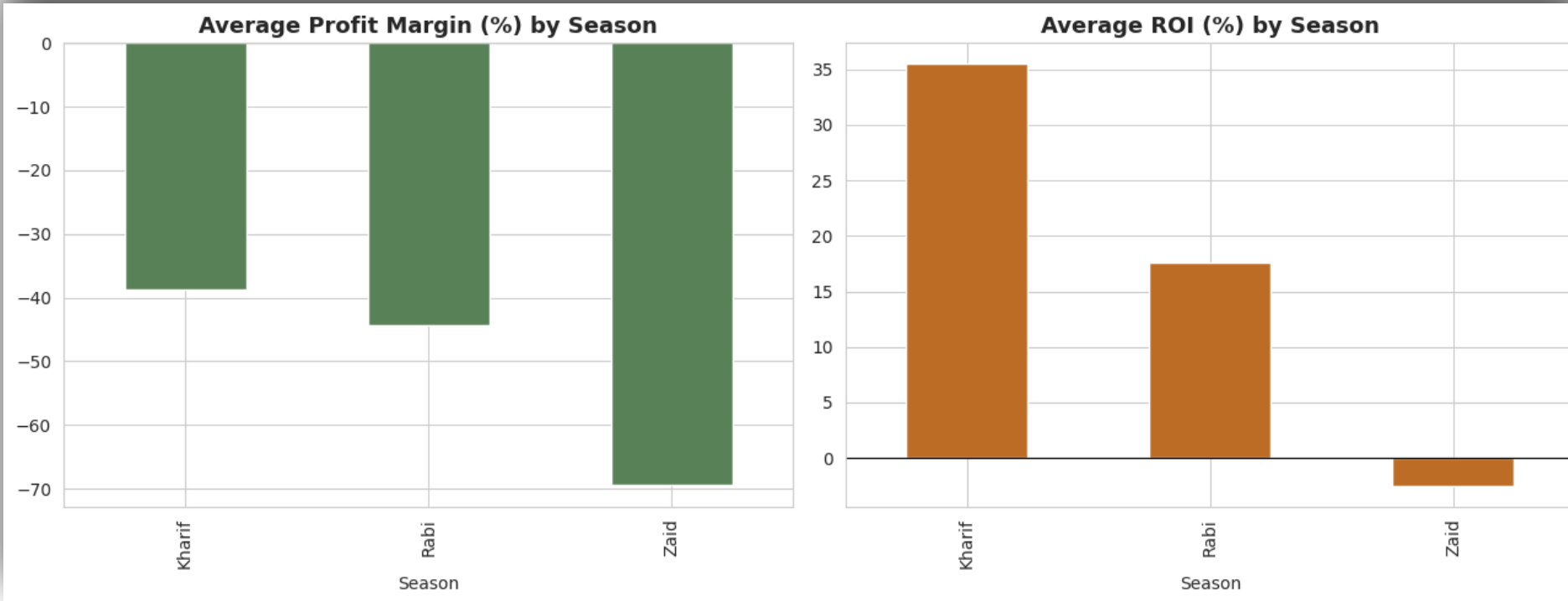
Rainfed irrigation is the most water-efficient method in every season (up to 5.63 t per 1000m³ in Kharif)

Flood irrigation is consistently the least efficient, regardless of season (as low as 2.21 in Zaid)

The gap between best and worst irrigation method is much larger than the gap between seasons – irrigation choice matters more than timing

```
irrig_season_pivot = df.pivot_table(values='Water_Efficiency_t_per_1000m3', index='Irrigation_Method', c
plt.figure(figsize=(7,5))
sns.heatmap(irrig_season_pivot, annot=True, fmt='.2f', cmap='PuBuGn')
plt.title('Water Efficiency: Irrigation Method vs Season')
plt.tight_layout()
plt.show()
```


RESULTS



Kharif has the strongest average ROI (~35%), followed by Rabi (~17.5%)

Zaid's average ROI is negative (~-3%) — confirming it's a loss-making season on average, not just occasionally

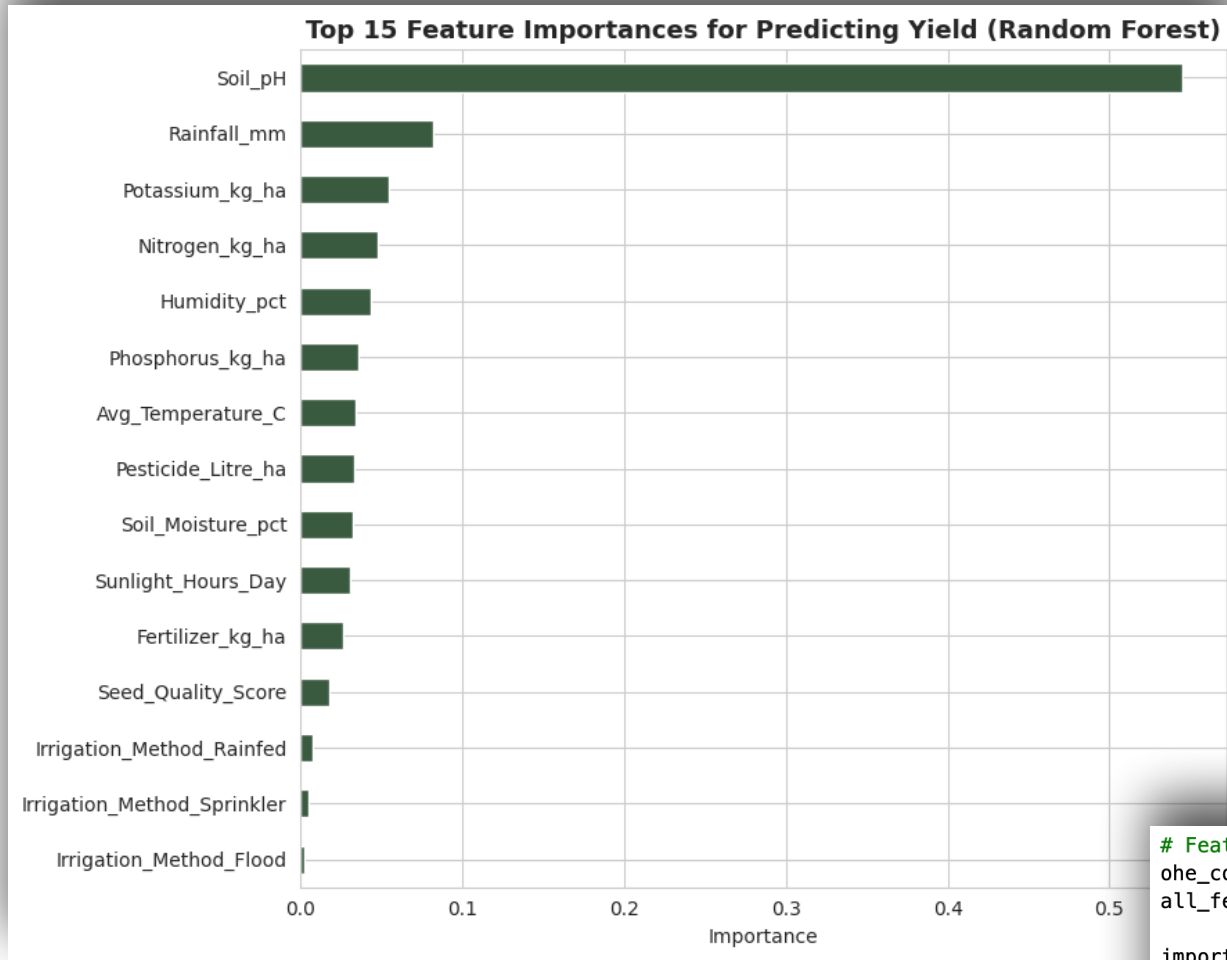
Average profit margin is negative across all three seasons, even Kharif — meaning a subset of farms with very low revenue and disproportionately high costs are dragging the *average* down.

```
fig, axes = plt.subplots(1, 2, figsize=(13,5))
efficiency_summary['Avg_Profit_Margin_pct'].plot(kind='bar', ax=axes[0], color='#588157')
axes[0].set_title('Average Profit Margin (%) by Season')
axes[0].axhline(0, color='black', linewidth=0.8)

efficiency_summary['Avg_ROI_pct'].plot(kind='bar', ax=axes[1], color='#BC6C25')
axes[1].set_title('Average ROI (%) by Season')
axes[1].axhline(0, color='black', linewidth=0.8)

plt.tight_layout()
plt.show()
```

RESULTS



- Soil pH is by far the strongest predictor of yield – its importance score (~0.55) dwarfs every other factor
- Rainfall, Potassium, and Nitrogen follow as the next most influential factors
- Irrigation method ranks lowest among all features – reinforcing that *what* you irrigate with matters less for yield prediction than soil and nutrient conditions (even though it mattered a lot for water efficiency in slide 3)

```
# Feature importance from Random Forest
ohe_cols = rf_pipeline.named_steps['prep'].named_transformers_['cat'].get_feature_names_out(categorical)
all_feature_names = list(ohe_cols) + numeric

importances = rf_pipeline.named_steps['model'].feature_importances_
feat_imp = pd.Series(importances, index=all_feature_names).sort_values(ascending=False)

plt.figure(figsize=(9,7))
feat_imp.head(15).plot(kind='barh', color='#3A5A40')
plt.gca().invert_yaxis()
plt.title('Top 15 Feature Importances for Predicting Yield (Random Forest)')
plt.xlabel('Importance')
plt.tight_layout()
plt.show()
```

Future scope

- Expand to multi-year data to validate whether seasonal patterns hold consistently over time
- Add variables like crop variety, farmer practice, and market prices to improve weak model performance (R^2 as low as 0.03-0.22)
- Analyze at district/village level for more localized insights
- Build an interactive dashboard for real-time yield/profit predictions by season and region
- Study the cost-benefit of farmers switching from Flood to Drip/Rainfed irrigation
- Incorporate climate-risk forecasting, especially for Kharif's higher disease/pest risk

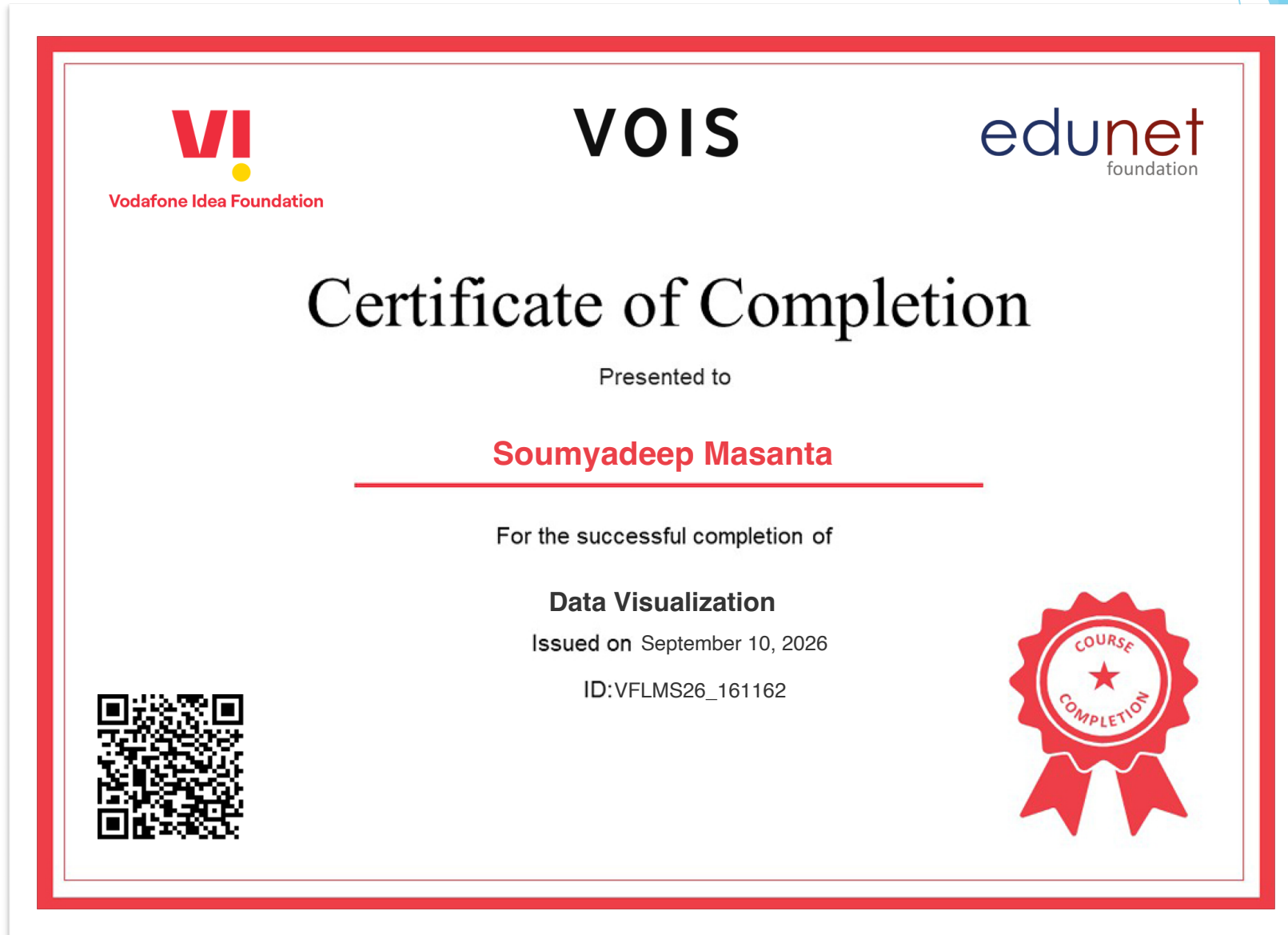
GitHub Link

- https://github.com/Soumyadeepmasanta/Seasonal_Agriculture_Performance_Analysis



VOIS Course completion certificate -

Course name- Data Visualization





Thank you